Problem Set # 1: Exploring Wage Differences in the U.S. Early Care and Education Workforce

In the United States, the early care and education (ECE) workforce is highly educated, with nearly 80% of educators having attended some college and about half holding an associate degree or higher (Coffey, 2022). Despite their educational attainment, ECE teachers earn substantially less than their kindergarten counterparts. According to the Center for the Study of Child Care Employment at UC Berkeley, a full-time early childhood teacher with a bachelor's degree earns only 57% of what a kindergarten teacher earns. In addition, real wages for ECE educators have declined over time, falling from $14.99 per hour in 2012 to $14.01 per hour in 2019 (Coffey, 2022).

Low wages contribute to high turnover rates in ECE programs, reaching as high as 30% in some settings (Caven et al., 2021). This instability creates financial hardship for educators while weakening the quality of early learning environments. Young children rely on consistent, responsive relationships with caregivers to support healthy development, and frequent staff turnover disrupts these critical connections (Caven et al., 2021). Turnover also places additional strain on remaining staff and program leadership, who must continually adjust to changing roles and team dynamics. Furthermore, it undermines quality improvement efforts by reducing the return on investments in professional development and shifting attention from educational support to ongoing staffing challenges (Caven et al., 2021).

Because ECE teachers play a critical role in promoting children's school readiness and long-term success, workforce stability is essential to maintaining high-quality early learning programs. These challenges underscore the need for policy solutions that improve compensation and strengthen support for ECE professionals.

Before developing policy solutions, however, it is important to understand the characteristics of the workforce experiencing low wages. Who earns the lowest wages? Are there wage differences across racial and ethnic groups or by gender? Are lower wages associated with educational attainment or years of experience? Understanding these patterns can help policymakers design interventions that target the populations most affected by low compensation.

In this assignment, you will use data from the National Survey of Early Care and Education (NSECE) 2019 to examine the characteristics of the ECE workforce and identify factors associated with lower wages. Please answer the following questions.

1. **Prepare the data for analysis.** Subset the NSECE dataset to include the following variables:
    - WF9_WORK_WAGE (hourly wage)
    - WF9_WORK_ROLE (staff role)

- WF9_WORK_YRS (number of years worked in the program)
- WF9_CHAR_HISP (Hispanic/Latino ethnicity)
- WF9_CHAR_RACE (race)
- WF9_CHAR_EDUC (highest level of education completed)
- WF9_CHAR_GENDER (gender)

Rename the variables using clear, descriptive names. Perform any data cleaning you consider necessary. For each variable, identify its level of measurement (nominal, ordinal, or interval/ratio) and whether it is discrete or continuous. Include the appropriate descriptive statistics, tables, or graphs, as well as measures of central tendency, dispersion, and shape when applicable. Describe and interpret your findings. **(4 points)**

2. **Examine the following relationships.** For each analysis, include the appropriate statistical test to determine whether the observed differences or associations are statistically significant. Interpret your results in the context of the ECE workforce. **(4 points)**
    - a. Hourly wage and staff role
    - b. Hourly wage and number of years worked in the program
    - c. Hourly wage and race/ethnicity
    - d. Hourly wage and educational attainment
    - e. Hourly wage and gender

3. **Policy implications.** Based on your analyses, describe the segment(s) of the ECE workforce that appear to be most affected by low wages. What characteristics would you consider when designing a policy to improve compensation in the ECE sector? Support your answer using evidence from your analyses. **(2 points)**